European Journal of Medical Genetics 61 (2018) 68–71

Contents lists available at ScienceDirect

European Journal of Medical Genetics

journal homepage: www.elsevier.com/locate/ejmg

Familial early-onset deep venous thrombosis associated with a novel *HRG* mutation

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A R T I C L E I N F O

Keywords:
Deep venous thrombosis
HRG mutation
Whole exome sequencing
Plasma protein deficiency

A B S T R A C T

Deep venous thrombosis (DVT) remains a serious clinical problem that affects millions of people worldwide. Some DVT cases are caused by inherited thrombophilia derived from genetic aberrations and several disease-causing genes have been identified so far. Among them, *HRG* is an uncommon one with limited related reports. Here, we reported on a family with early-onset DVT where acquired risky conditions were excluded. Whole exome sequencing revealed a novel heterozygous single base pair substitution in exon 2 of *HRG* gene resulting in a conserved residue replacement of the protein (c. C271T, p. P73S). Sanger sequencing confirmed the co-segregation of the mutation and plasma quantification determined circulating protein deficiency. The mutation might therefore impair hemostatic balance by causing reduced circulating HRG level. Our study broadens the mutation spectrum of the *HRG* gene and underscores the importance of its function in regulating coagulation pathway.

1. Introduction

Deep venous thrombosis (DVT) represents a significant clinical problem that affects 1 in 1000 individuals annually and can results in fatal pulmonary embolism (Cushman, 2007). Despite given standard anticoagulant therapy, roughly 25-50% of patients with DVT will eventually develop post-thrombotic syndrome (PTS) characterized by chronic swelling, itching, and ultimately ulceration (Prandoni et al., 2016). Various risk factors such as surgery, trauma, cancer, pregnancy, and long-term immobilization could predispose to DVT. Besides acquired conditions, inherited thrombophilia should also be considered, especially in patients with positive family history and early-onset episodes (Yilmaz and Gunaydin, 2015).

Well recognized genetic disorders causing inherited thrombophilia included prothrombin mutation, factor V Leiden mutation, and mutations in genes encoding antithrombin, protein C, or protein S (Miyata et al., 2016). Histidine-rich glycoprotein (HRG) is an abundant plasma glycoprotein that exerts multiple biological functions (Jones et al., 2005). It could mediate hemostatic balance via binding to a variety of ligands such as heparin, heparan sulfate, thrombospondin, and plasminogen. The *HRG* gene (MIM, 142640) extends over ∼18 kb of genomic DNA and contains 7 exons that generate a 2 kb mRNA. Plasma deficiency of HRG caused by gene mutations have been reported to be responsible for thrombophilia pedigrees (Shigekiyo et al., 1998, 2000). However, its mutation profile and clinical features concerning HRG are poorly characterized due to limited related reports, thus necessitating further research.

Here, we described the case of a man who developed early-onset DVT without acquired predisposing risk factors. Multiple family members had also been affected. The identification of a strong risk factor for thrombosis (a novel *HRG* mutation with damaging effect) combined with quantity analysis of plasma HRG might explain the clinical phenotype.

2. Materials and methods

2.1. Subjects

A 41 year-old man was referred to our department complaining of repeated left lower extremity swelling and itching. Physical examination revealed brownish skin discoloration around the ankle. At the age of 19, he had progressive pain and swelling of the left leg. He denied history of trauma or immobilization before the episode. Diagnosis of PTS was suspected and deep venous angiography was performed. Angiography images showed occlusion of deep veins accompanied by marked collateral vessel formation (Fig. 1A). Coagulation screening

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http://dx.doi.org/10.1016/j.ejmg.2017.10.019
Received 12 June 2017; Received in revised form 29 September 2017; Accepted 29 October 2017
Available online 07 November 2017
1769-7212/

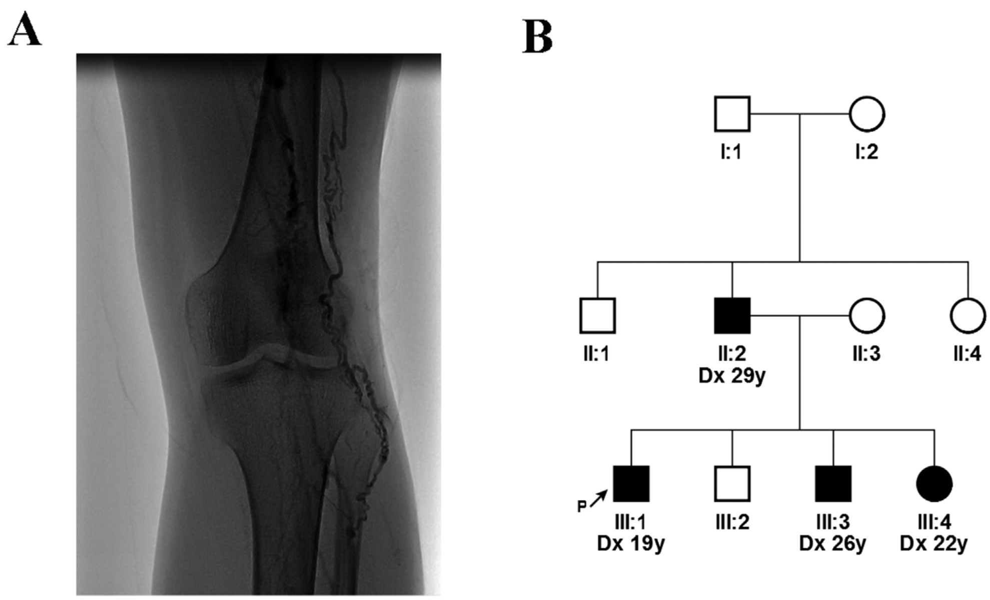

Fig. 1. Radiographic findings and pedigree. (A) Deep venous angiography demonstrated occlusion of deep veins accompanied by marked collateral vessel formation. (B) Proband is indicated with an arrow. Fully filled symbols, DVT; unfilled symbols, unaffected. The age at diagnosis of DVT ("dx") is shown in years.

Table 1
Clinical characteristics and blood results of family members.

	II:1	II:2	II:4	III:1	III:2	III:3	III:4
Age (years)	66	64	60	41	38	34	30
Gender	M	M	F	M	M	M	F
DVT	–	+	–	+	–	+	+
Age at diagnosis (years)	NA	29	NA	19	NA	26	22
PT (s)	11.2	10.1	10.7	11.3	11.5	11.7	11.3
APTT (s)	28.5	25.3	22.6	25.0	32.8	33.4	27.6
Fibrinogen (g/L)	3.20	2.95	3.65	3.38	2.51	2.54	3.02
Plasma HRG level (ug/ml)[a]	102.6	50.1	117.5	60.6	118.3	59.2	61.3
HRG mutation	–	+	–	+	–	+	+

DVT, deep venous thrombosis; PT, prothrombin time; APTT, activated partial thromboplastin time; HRG, histidine-rich glycoprotein; M, male; F, female; NA, not applicable.
[a] Normal range is 101.2–127.7 μg/ml derived from 10 control samples in this study.

tests revealed no abnormalities (Table 1). Plasma level of antithrombin activity, protein C activity and protein S activity were within the normal limit. Concentration of antiphospholipid antibodies were below the respective cut-off values, and screening tests for the lupus anticoagulant were negative. His family history was positive for DVT episodes, with his father and siblings diagnosed at the age of 29, 26 and 22, respectively (Fig. 1B). A total of 200 unrelated ethnically matched normal controls were recruited from the same region. The study complied with the Declaration of Helsinki and was approved by the local Ethics committee. After written informed consents were obtained from all participants, venous blood samples were collected and genomic DNA were extracted per protocol using the QIAamp DNA Blood Mini Kit (Qiagen, Hilden, German).

2.2. Exome capture

To understand the genetic predisposition to thrombophilia of the pedigree, whole exome sequencing (WES) of the proband (III:1) was performed using the Agilent SureSelect Human All Exon V6 Kit on the Illumina HiSeq 2500 platform by Novogene Bioinformatics Institute, Beijing, China. A minimum of 1.5 μg of genomic DNA was utilized to construct the exome library. The genomic DNA was sheared into 180-280bp by sonication and hybridized for enrichment, according to the manufacturer's guideline. Enriched DNA was sequenced on the Illumina HiSeq 2500 platform to generate paired-end reads with length of 150bp.

2.3. Variant identification and analysis

The sequence reads were mapped to the human reference genome (UCSC hg19) with the Burrows-Wheeler Aligner (BWA). Picard (http://sourceforge.net/projects/picard/) was utilized to mark duplicate reads derived from PCR amplification. SAMtools was employed to do variant calling and identify single nucleotide variants (SNVs) or small insertions and deletions (Indels). The called SNVs and Indels were annotated with ANNOVAR. All candidate variants were filtered against the dbSNP (http://www.ncbi.nlm.nih.gov/projects/SNP/), 1000G (http://www.1000genomes.org/), ESP (http://evs.gs.washington.edu/EVS), and ExAC (http://exac.broadinstitute.org) to remove the polymorphism loci. The PolyPhen2 algorithm (http://genetics.bwh.harvard.edu/pph2/) was performed to predict whether an amino acid substitution affects protein function. Amino acid conservation across species was evaluated by GERP score. Sanger sequencing was performed to validate the identified potential disease-causing variant in family members and 200 normal controls.

2.4. Evaluation of HRG level

Plasma was separated from the blood samples and frozen at -20 °C. The HRG levels in the plasma were analyzed using the commercially available ELISA kits (Sino Biol, Beijing, China) according to the manufacturer's instructions. The samples were diluted when needed. Absolute plasma concentrations for HRG are given in ug/ml.

3. Results

An overview of quality assessment of the exome sequencing is given in Table 2. Exome sequencing yielded a total number of 24,192 variants

Table 2
Quality assessement of exome sequencing data of the proband.

Raw data	10.94Gb
Data passed filtering	98.48%
Data with Q score > 30	89.52%
Data mapped	98.21%
Fraction of target with at least 20X	87.20%
Average sequencing depth on target	113X
Total number of SNVs identified in exonic or splicing region	23245
Total number of Indels in exonic or splicing region	947

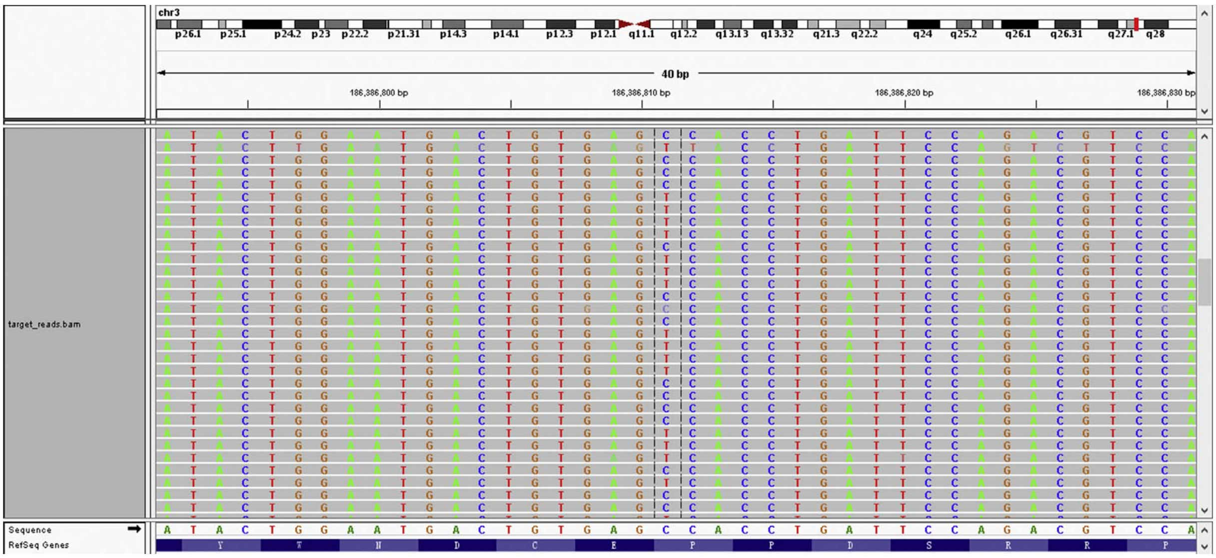

Fig. 2. Pictorial representation of the chr3: 186386811 C > T mutation (c.271C > T) identified in *HRG* gene through Integrative Genomics Viewer (IGV).

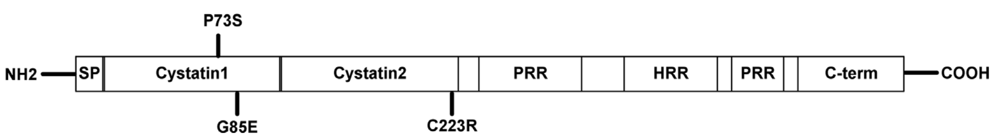

Fig. 3. Schematic diagram of the HRG protein and the locations of the 3 non-synonymous mutations identified so far. Mutations identified in this study are shown above the protein structure; mutations reported in previous studies are underneath. SP, signal peptide; PRR, proline-rich region; HRR, histidine-rich region.

located in the exonic or spicing region. Following standard filtering procedure as previously described, we identified 289 suspicious variants, including 261 SNVs and 28 Indels (Zhang et al., 2017). We focused on the variants located in the thrombophilia candidate genes or genes implicated in blood coagulation process. Among these genes, *HRG* (NM_000412) was the only remaining candidate gene for causing DVT because of its established role in thrombosis and published literature demonstrating that *HRG* mutations predispose to thrombophilia. A missense variant, c.271C > T (p. Pro73Ser) in exon 2 of *HRG* was identified. For this site, 49% (78/159) reads supported for C, whereas 51% (81/159) reads supported for T (Fig. 2). The *HRG* variant falls in the second cystatin-like homology domain and is predicted to be damaging by Polyphen2 (Fig. 3). This altered residue also appears evolutionarily conserved with GERP score of 3.52. It is not found in 13,000 chromosomes in the ESP database and only present twice in 121,214 chromosomes, i.e., MAF 1.65×10^{-5} in the ExAC database. Variant accession for the variant has been deposited in ClinVar (accession number SCV000605916).

Sanger sequencing confirmed the variant in the proband and affected relatives (II:2, III:3 and III:4), suggesting that it co-segregates with DVT in the family (Table 1). Additionally, the variant was not present in 200 unrelated ethnically matched normal controls. Quantity analysis of plasma HRG revealed approximately half concentration of affected individuals compared with those unaffected (mean value: 57.8 μg/ml vs 112.8 μg/ml), suggesting that haploinsufficiency might underlie the pathogenic role of this mutation. Collectively, our analysis supports that it is a pathogenic mutation that accounts for DVT episodes in this family.

4. Discussion

Inherited thrombophilia is defined as a genetically determined predisposition to venous thromboembolism which characteristically occurs at a relatively young age without apparent causes and tends to recur (Reitsma, 2015). Antithrombin deficiency was the first identified genetic risk factor for thrombophilia and more coagulation-related genes have been unraveled in the past decades. Identification of mutations in certain disease-causing genes not only aids in ultimate diagnosis of thrombophilia patients with unknown etiology, but also contributes to individualized therapy. Furthermore, we could achieve a better understanding of coagulation pathway and pathogenic mechanism by exploring the profile of mutated genes regarding thrombophilia. The proband in this study developed early-onset DVT without acquired known risk factors. Positive family history prompted us to investigate the underlying genetic causes. WES has emerged as a robust technique to identify disease-causing mutations (Ng et al., 2010). By performing WES, we revealed a novel missense mutation of *HRG*, which is one of the disease-causing genes for thrombophilia. Population variation data, bioinformatics analysis and protein quantification all support its pathogenic role.

HRG is a plasma adaptor protein that is present at a relatively high concentration in adult blood (Jones et al., 2005). It has the capacity to bind with a strikingly large number of ligands via its multi-domain structure to regulate numerous important biologic processes, such as coagulation, fibrinolysis, immune complex clearance, cell adhesion, and angiogenesis (Poon et al., 2011). The multi-domain structure of HRG comprises of two cystatin-like homology domains, a central histidine-rich region (HRR) flanked by 2 proline-rich regions (PRR1 and PRR2), and a C-terminal domain. The precise function HRG played is ligand-dependent and hemostatic property is largely attributed to the association with heparin (Lijnen et al., 1983). HRG binds heparin with high affinity and has been reported to neutralize the anticoagulant activity of heparin by inhibiting the formation of heparin-antithrombin III complexes. Moreover, the presence of Zn ion released by activated platelets can augment the interaction between HRG and heparin, and potentiate the ability of HRG to neutralize heparin (Fu and Horn, 2002). The two cystatin-like homology domains serve as major bind sites with heparin and disruption of such domains might result in coagulation disorders. Aligned with the observation, *HRG* mutations currently found in thrombophilia patients are all located in this region (Table 3). Besides venous thrombosis, HRG has also been implicated in arterial thrombosis, as observed in HRG-deficient mice (Vu et al.,

Table 3
Overview of currently reported *HRG* mutations so far.

c-notation	p-notation	Exon	Domain	Diseases	Function	Reference
c.271C > T	p.Pro73Ser	2	Cystatin 1	Deep venous thrombosis	Plasma protein deficiency	Current study
c.308G > A	p.Gly85Glu	3	Cystatin 1	Dual arteriovenous fistula	Plasma protein deficiency	Shigekiyo et al., 1998
c.721 T > C	p.Cys223Arg	6	Cystatin 2	Cerebral sinus thrombosis	Plasma protein deficiency	Shigekiyo et al., 2000

The *HRG* reference sequence used was NM_000412.4, in which the A of the ATG translation initiation codon was designtaed nucleotide 1.

2016). HRG has the capacity to inhibit thrombosis by modulating the intrinsic pathway of coagulation.

Mutations identified in the *HRG* gene are all missense mutation that caused deficiency of plasma HRG. Wakabayashi et al. expressed the *HRG* mutants (G85E and C223R) in baby hamster kidney cells to explore the mechanism and molecular bases of plasma HRG deficiency (Wakabayashi et al., 2000). They found that both mutants were only partially secreted and mainly degraded within the cells. Haploinsufficiency is therefore proposed as a pathogenic mechanism underlying *HRG* mutations. It is noteworthy that multiple thrombophilia families had elevated levels of plasma HRG, as reported in several studies (Castaman et al., 1993). This seemed contradictory to the phenomenon observed in *HRG*-mutated families. One possible explanation is that *HRG* mutations lead to decreased levels of plasma HRG, whereas *HRG* mutations negative patients have compensatory elevated plasma secondary to thromboembolism events. The spectrum of coagulation disorders currently reported in *HRG* mutation positive patients includes transverse sinus thrombosis, dual arteriovenous fistula at sites of previous thrombosis and DVT present in this case. A previous report had demonstrated a family case with decreased plasma levels of HRG in 4 individuals of 3 generations (Souto et al., 1996). The proband developed pulmonary embolism twice at 36, while his father had central retinal artery thrombosis at 59. Although genetic screening of *HRG* was not performed, *HRG* mutation was highly likely to be responsible for their clinical manifestation.

In conclusion, we performed whole-exome and direct sequencing to uncover the genetic cause of early-onset DVT in a Chinese family with multiple members affected and identified a novel mutation (c. C271T, p. P73S) in the *HRG* gene. The mutation might impair hemostatic balance by causing reduced circulating HRG level. Our study broadens the mutation spectrum of the *HRG* gene and underscores the importance of its function in regulating coagulation process.

Disclosure

The authors declare no conflict of interest.

Acknowledgements

We would like to thank the family for their participation in this study. This work was supported by the grants from National Natural Science Foundation of China (no. 81660086).

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